CNENGR 116
Fundamentals of Digital Electronics

Lecture 5 – Karnaugh Maps

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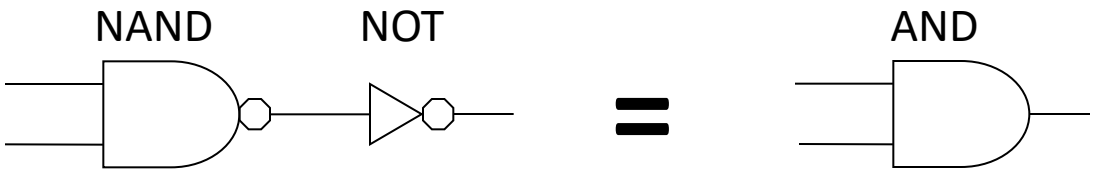
Revision: Combinational logic

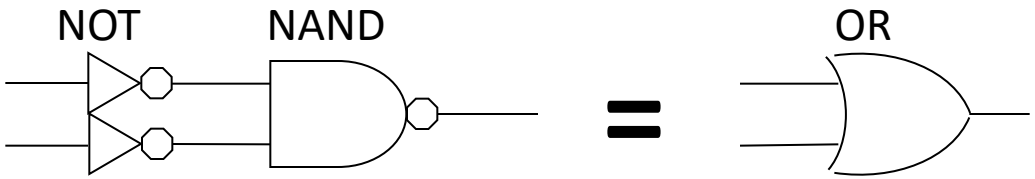
- Combinational logic circuits
  - Respond to incoming data (0's and 1's) almost immediately and without regard to earlier events (this will make more sense when we learn about sequential circuits)
  - Virtually ANY combinational circuit can be implemented with only "NAND" or "NOR" gates

Revision: Combinational logic

- AND gates made from NAND and NOT gates

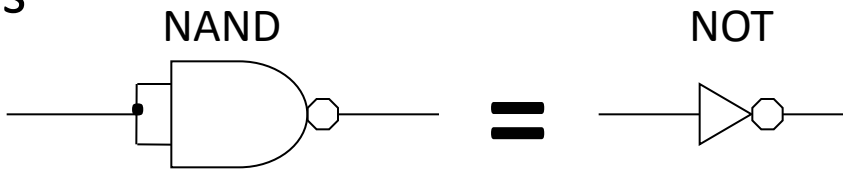


- OR gates made from NAND and NOT gates



$$\overline{A\bar{A}} = \bar{A} + \bar{A} = \bar{A}$$

- NOT gates made from NAND gates



Revision: Standard SOP form

- A **standard SOP** expression – ALL the variables in the domain appear in each product term in the expression.

  Example: $A\overline{B}CD + \overline{A}\,\overline{B}C\overline{D} + AB\overline{C}\,\overline{D}$

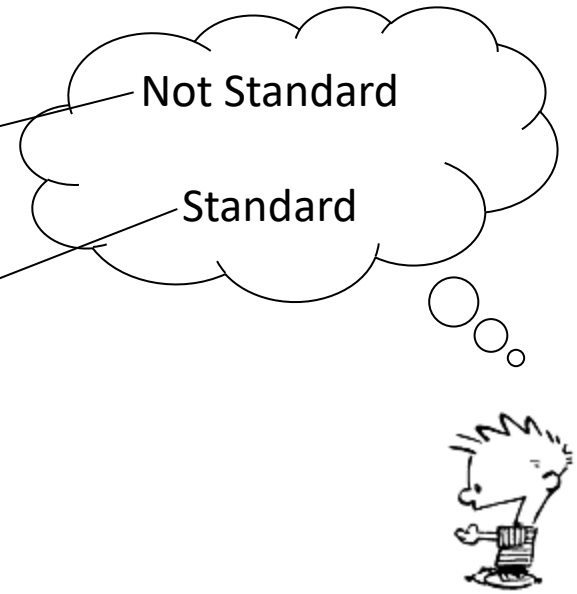
- Standard SOP expressions are important for construction of truth tables and in the Karnaugh map simplification method.

Revision: Standard POS form

- A **standard POS** – each sum term contains ALL the variables for the domain of the expression.
- Examples:

$$(A+\overline{B}+C)(A+B+\overline{D})(A+\overline{B}+\overline{C}+D)$$

$$(\overline{A}+\overline{B}+\overline{C}+\overline{D})(A+\overline{B}+C+D)(A+B+\overline{C}+D)$$



Revision: Converting to Standard SOP and to Standard POS

- **Convert to standard SOP**
  - $\times(A + \bar{A})$ This is rule #6 $(A + \bar{A} = 1)$
  - Then multiply out
- **Convert to standard POS**
  - $+A\bar{A}$ This is rule #8 $(A\bar{A} = 0)$
  - Then use rule #12 $(A + B)(A + C) = A + BC$

Revision: Converting standard POS to standard SOP

- The binary value of the products terms in a given standard SOP expression are not present in the equivalent standard POS expression. And the same goes for POS values in SOP expressions.
- Therefore to convert from SOP to POS follow these steps:
  1. Determine the binary numbers that represent each product term in SOP expression
  2. Determine all the binary numbers that are NOT included in step 1
  3. Write equivalent sum term for each binary number from step 2, and express in POS form

Revision: Example of converting standard POS to standard SOP

- Convert the following SOP to equivalent POS:

$$\overline{A}\,\overline{B}\,\overline{C} + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}C + ABC$$

What must A, B & C equal for the expression to equal 1

- Binary numbers are: 000+010+011+101+111
- Since there are 3 variables in the domain then there are 8 = 2^3 possible combinations. The SOP expression contain 5 of these combinations, so the POS must contain the other 3 which are 001, 100, 110:

$$(A + B + \overline{C})(\overline{A} + B + C)(\overline{A} + \overline{B} + C)$$

What must A, B & C equal for the expression to equal 0

Revision: Conversion of SOP to Truth table

SOP expression is equal to 1 only if at least one of the product terms equal to 1.

1. Construct a truth table with all possible values of the expression.

2. Convert SOP into standard form and find binary values for each term.

3. Place 1 in the truth table in a row corresponding to each of these binary values. The rest of the rows are zero.

Example:

Construct truth table for the Standard SOP expression: $\overline{A}\overline{B}C + A\overline{B}\overline{C} + ABC$

$\overline{A}\overline{B}C : 001; A\overline{B}\overline{C} : 100; ABC : 111$

Inputs			Output	
A	**B**	**C**	**X**	**Product Term**
0	0	0	0	
0	0	1	1	$\overline{A}\overline{B}C$
0	1	0	0	
0	1	1	0	
1	0	0	1	$A\overline{B}\overline{C}$
1	0	1	0	
1	1	0	0	
1	1	1	1	ABC

Overview

- Introduction to 3, 4 and 5 variable Karnaugh Maps
- Minimisation of SOP by means of Karnaugh maps
- Mapping truth tables directly onto Karnaugh maps

Good References

- Digital Fundamentals by Thomas L. Floyd

  (ISBN 10: 0131946099) basic text
- Principles of Digital Design by Daniel Gajski

  (Prentice Hall ISBN 0-13-301144-5) – advanced text

Objectives

- Introduce Karnaugh maps for 3, 4 and 5 variables
- Describe how to construct a Karnaugh map for a POS or a SOP
- Introduce a method of minimisation of a POS and SOP expressions by means of grouping cells in Karnaugh maps

Karnaugh map

- A Karnaugh map is similar to a truth table because it represents all possible values of input variables and the resulting output for each value.
- A Karnaugh map also provides a systematic method for simplifying Boolean expressions and if used correctly, will produce the simplest SOP or POS expressions possible.
- Instead of being organised into columns and rows like a truth table, the Karnaugh map is an array of cells in which each cell represents a binary value of the input variables.

Variables in Karnaugh map

- A Karnaugh map can be used for expressions with 2, 3, 4 and 5 variables. We will only cover 2 and 3 variable maps in this module.
- The number of cells in a Karnaugh map is equal to the total number of possible input variables. (3 variable => $2^3 = 8$).

3-variable Karnaugh map

Note the sequence of numbers

- Variables A, B and C
- A and B values are displayed on the left side of the Karnaugh map and C on the top.
- Combining AB and C in the appropriate cells effectively form a truth table.

AB \ C	0	1
00		
01		
11		
10		

Only one variable transition to move to neighbouring cell

3 and 4 - variable Karnaugh map

AB \ C	0	1
00	$\bar{A}\bar{B}\bar{C}$	$\bar{A}\bar{B}C$
01	$\bar{A}B\bar{C}$	$\bar{A}BC$
11	$AB\bar{C}$	ABC
10	$A\bar{B}\bar{C}$	$A\bar{B}C$

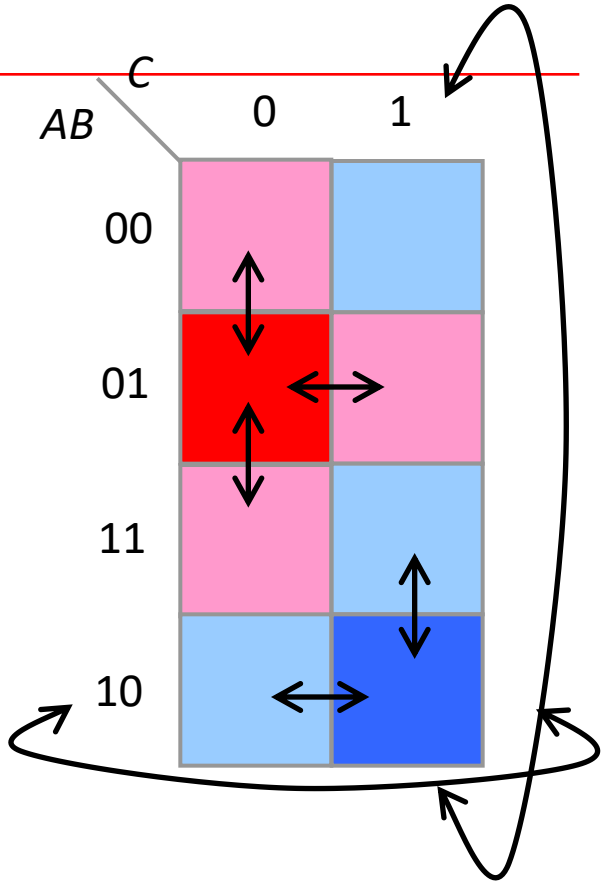
3 variable Karnaugh map

AB \ CD	00	01	11	10
00	$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}CD$	$\bar{A}\bar{B}C\bar{D}$
01	$\bar{A}B\bar{C}\bar{D}$	$\bar{A}B\bar{C}D$	$\bar{A}BCD$	$\bar{A}BC\bar{D}$
11	$AB\bar{C}\bar{D}$	$AB\bar{C}D$	$ABCD$	$ABC\bar{D}$
10	$A\bar{B}\bar{C}\bar{D}$	$A\bar{B}\bar{C}D$	$A\bar{B}CD$	$A\bar{B}C\bar{D}$

4 variable Karnaugh map

Cell Adjacency

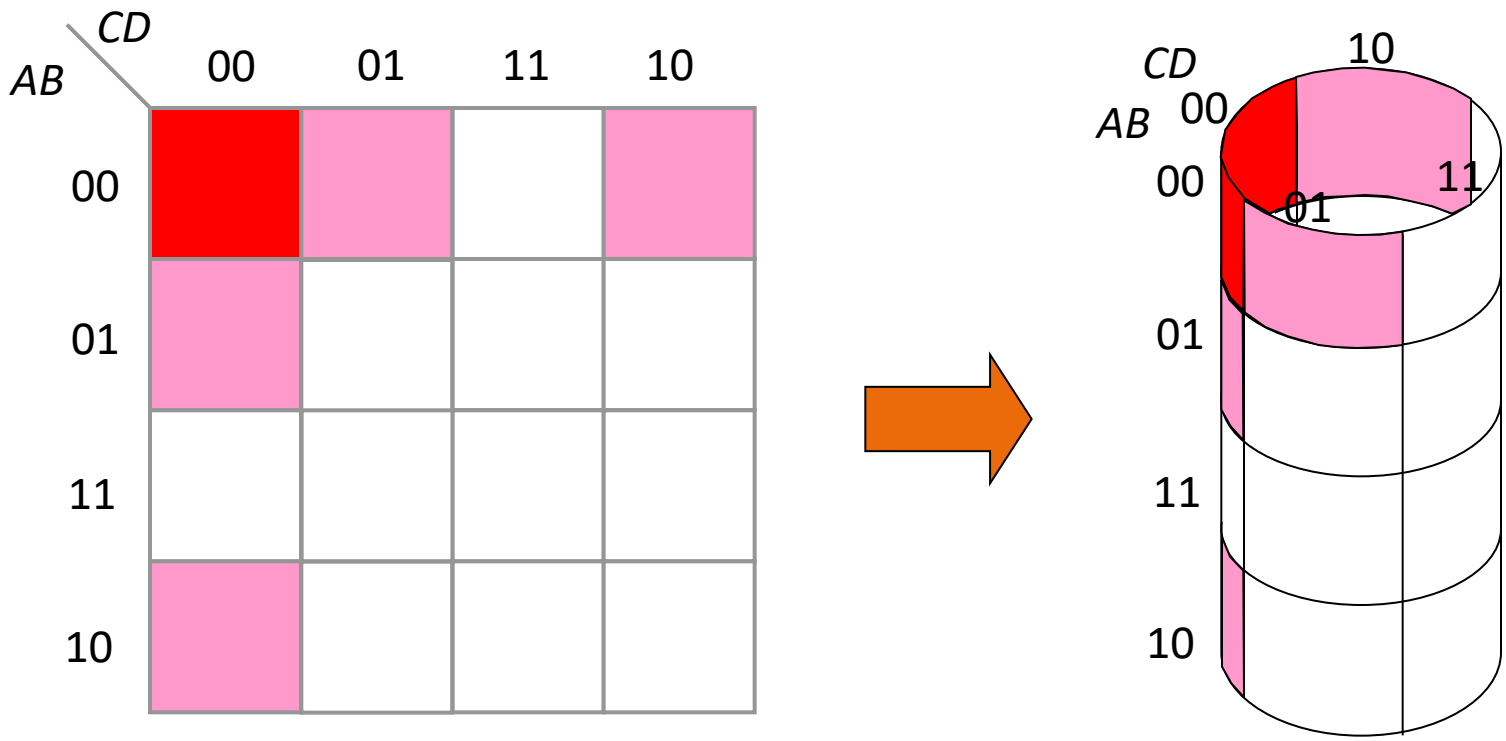
- The cells in the Karnaugh map are arranged so that there is only a single-variable change between adjacent cells.
- Example:
  - cell 010 is adjacent to 000, 011 and 110
  - cell 010 is NOT adjacent to 001, 111, 100 or 101
- Note that edge cells are adjacent to cells in the opposite edge.
- Example:
  - 101 is adjacent to 001 as well as 100 and 111.



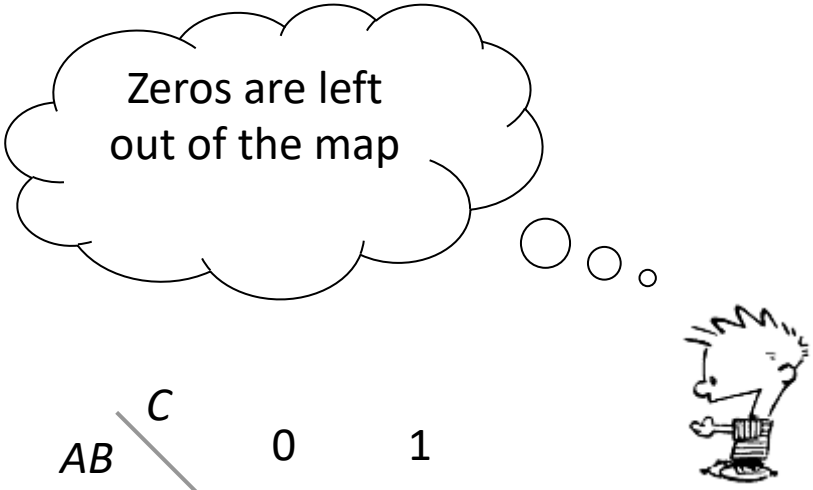
3 variable Karnaugh map

Cell Adjacency (expanded)

- In a 4 variable Karnaugh map adjacency is more complicated, but the same rules apply.
- One could think of a Karnaugh map as being a continuous cylinder (or a kind of cylinder) with <u>all</u> edges linking together.



Mapping a Standard SOP



- Each "1" is placed in a cell corresponding to a SOP term.
- For example, for a term $A\overline{B}C$ a "1" goes to the cell 101 in a 3-variable diagram.

AB \ C	0	1
00	1	
01		
11	1	
10	1	1

$$\overline{A}\,\overline{B}\,\overline{C} + A\overline{B}C + AB\overline{C} + A\overline{B}\,\overline{C}$$

Have a go at filling in this Karnaugh map…

Example: Mapping a Standard SOP on 3-variable K-map

- Map the following expression as shown below:

$$\overline{A}\,\overline{B}C + \overline{A}B\overline{C} + AB\overline{C} + ABC$$

001 010 110 111

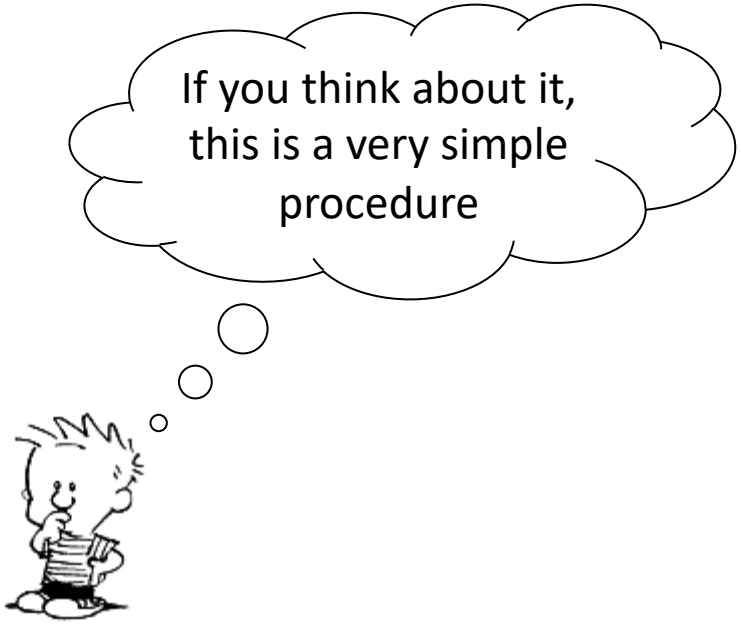
- Start by finding a binary representation of the SOP

AB \ C	0	1
00		1
01	1	
11	1	1
10		

Example: Mapping a Standard SOP on 4-variable K-map

$$\overline{A}\,\overline{B}CD + \overline{A}B\overline{C}\,\overline{D} + AB\overline{C}D + ABCD + AB\overline{C}\,\overline{D} + \overline{A}\,\overline{B}\,\overline{C}D + A\overline{B}C\overline{D}$$

0011 0100 1101 1111 1100 0001 1010



AB \ CD	00	01	11	10
00		1	1	
01	1			
11	1	1	1	
10				1

Example: Mapping a Non-Standard SOP on 3-variable K-map

- Map the following expression:

$$\overline{A} + A\overline{B} + AB\overline{C}$$

$$\downarrow \qquad \downarrow \qquad \downarrow$$

$\overline{A}\,\overline{B}\,\overline{C}$	$A\overline{B}\,\overline{C}$	$AB\overline{C}$
$\overline{A}\,\overline{B}C$	$A\overline{B}C$	
$\overline{A}B\overline{C}$		
$\overline{A}BC$		

$$\overline{A} + A\overline{B} + AB\overline{C}$$

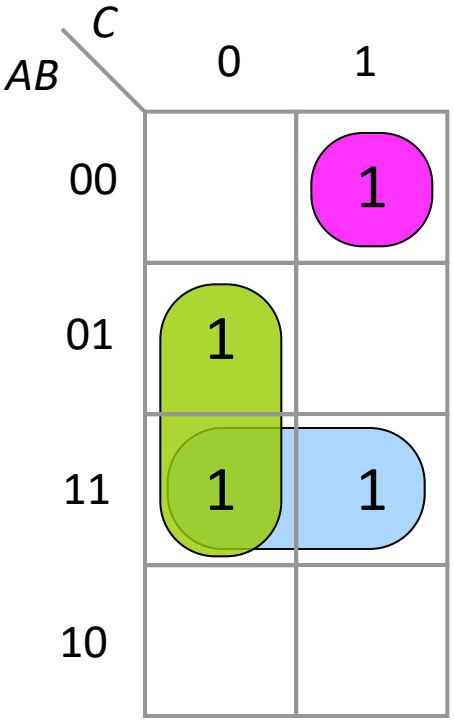
$$\downarrow \qquad \downarrow \qquad \downarrow$$

000	100	110
001	101	
010		
011		

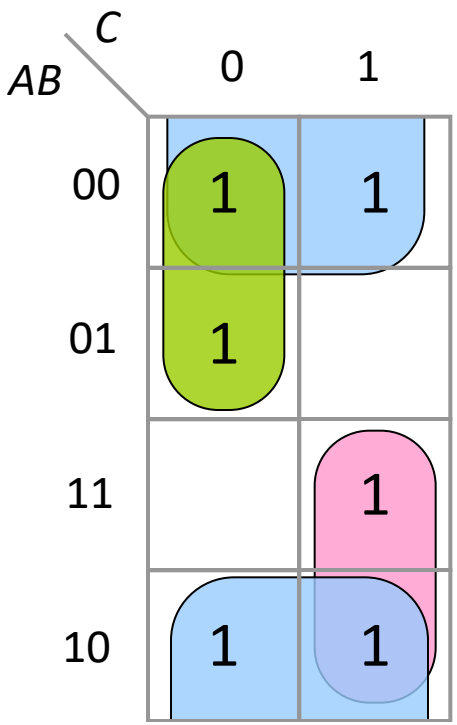
AB \ C	0	1
00	1	1
01	1	1
11	1	
10	1	1

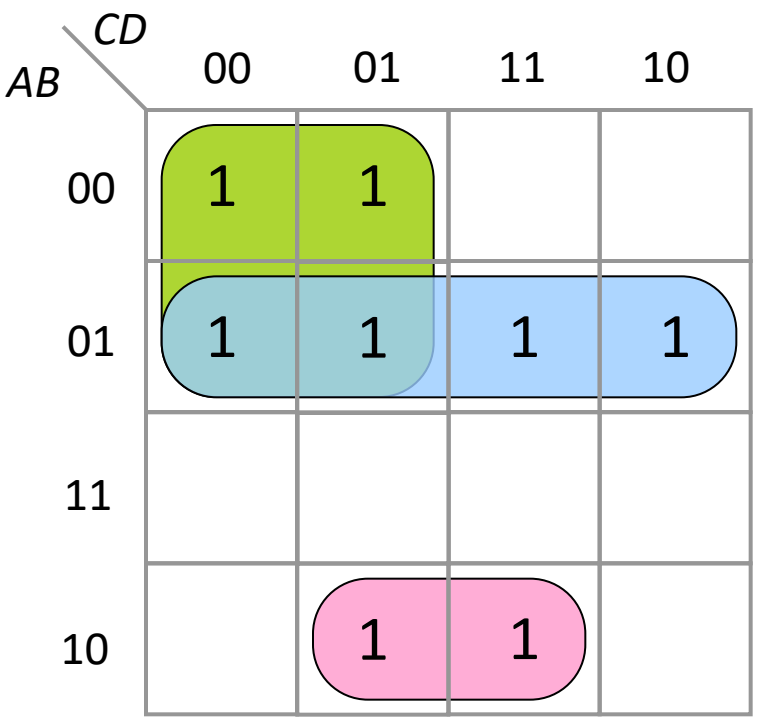
Karnaugh map simplification of SOP expressions

- We are now asked to do the reverse process and to simplify a SOP expression (to minimise it)
- This is done by grouping the “1” cells in the Karnaugh map according to the following rules:
  - A group of cells **must** contain 1, 2, 4, 8 or 16 cells
  - Each cell in a group must be adjacent to one or more cells in that group
  - **Always** include the largest possible number of cells in a group
  - Each “1” in the map **must be included in at least one group**
  - The “1s” already in a group **can** be included in another group as long as the groups are not identical



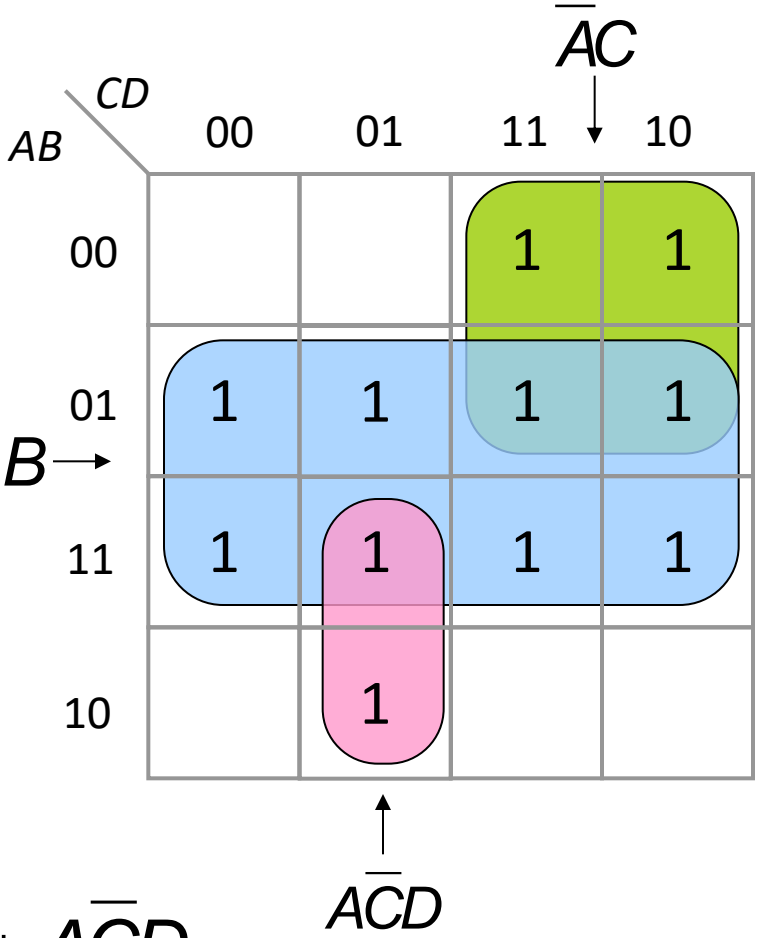
Examples of SOP groupings





Example: Find a minimum SOP expression

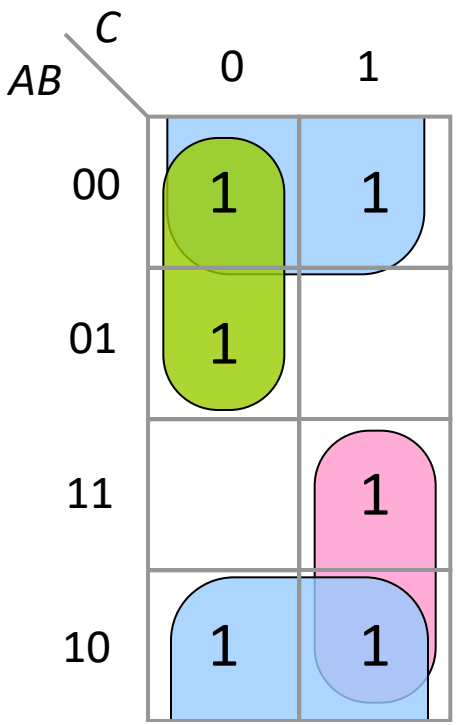
- Rule: **Eliminate variables that are in the same group in both complemented and uncomplemented forms.**
- Group of 8 cells:
  - A, C and D are all present therefore only B remains.
- Groups of 4 cells:
  - B and D are present therefore only $\overline{A}C$ remains.
- Group of 2 cells:
  - B is the only one present with its own complement therefore $A\overline{C}D$



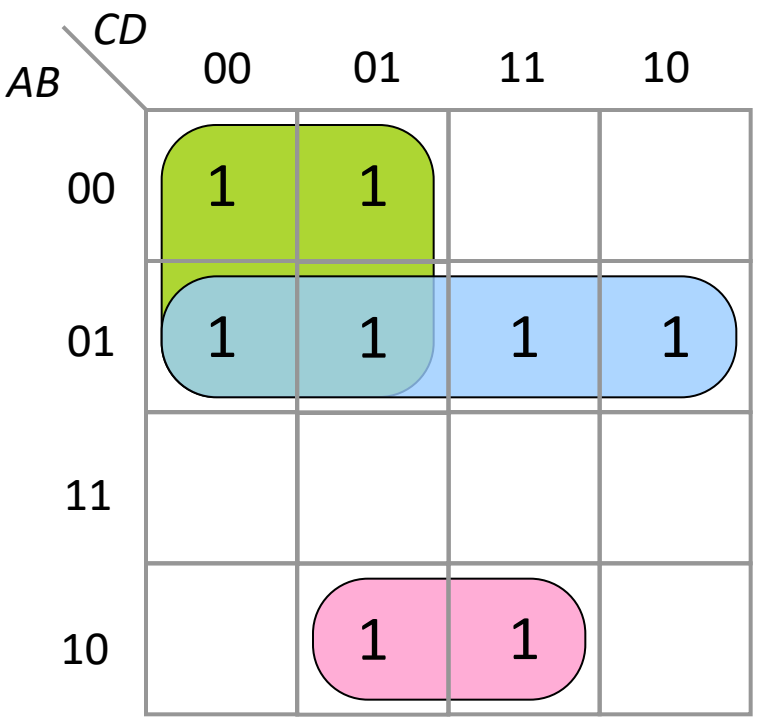
$$B + \overline{A}C + A\overline{C}D$$

Examples of SOP groupings

Have a go!



$$\bar{A}\bar{C} + \bar{B} + AC$$



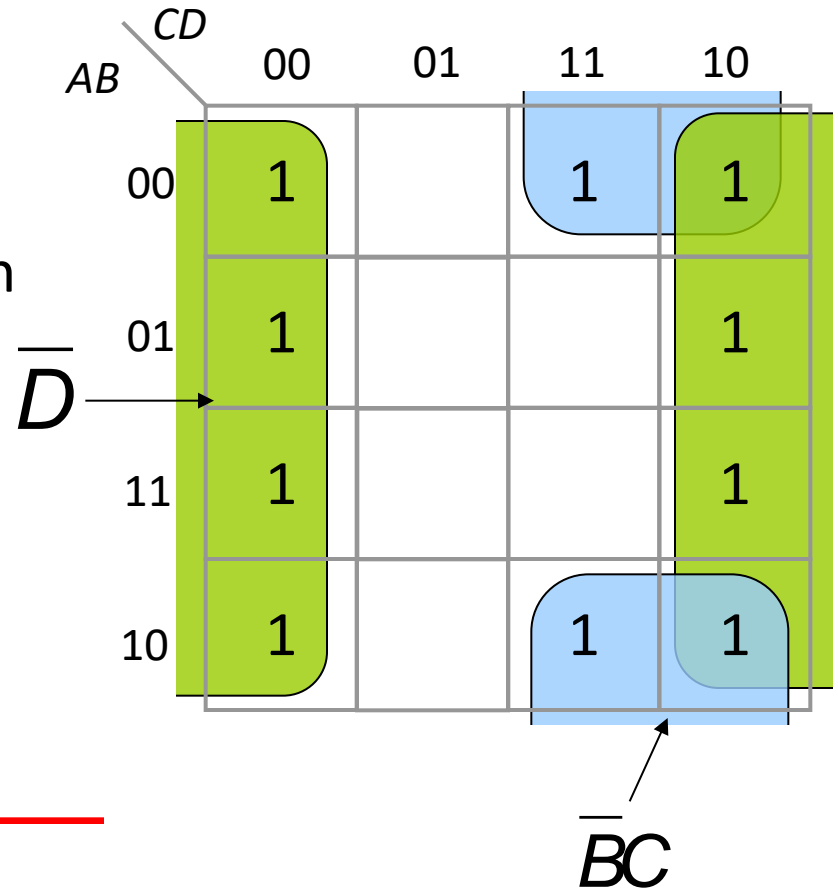
$$\bar{A}\bar{C} + \bar{A}B + A\bar{B}D$$

Example: SOP minimisation

$$\overline{B}\,\overline{C}\,\overline{D} + \overline{A}B\overline{C}\,\overline{D} + AB\overline{C}\,\overline{D} + \overline{A}\,\overline{B}CD + A\overline{B}CD + \overline{A}\,\overline{B}C\overline{D} + \overline{A}BC\overline{D} + ABC\overline{D} + A\overline{B}C\overline{D}$$

1. First convert term into standard form
2. Plot the expression on Karnaugh map
3. Group the "1s"
4. Find the answer

$$\overline{D} + \overline{B}C$$



Quick Exercise: Group the following

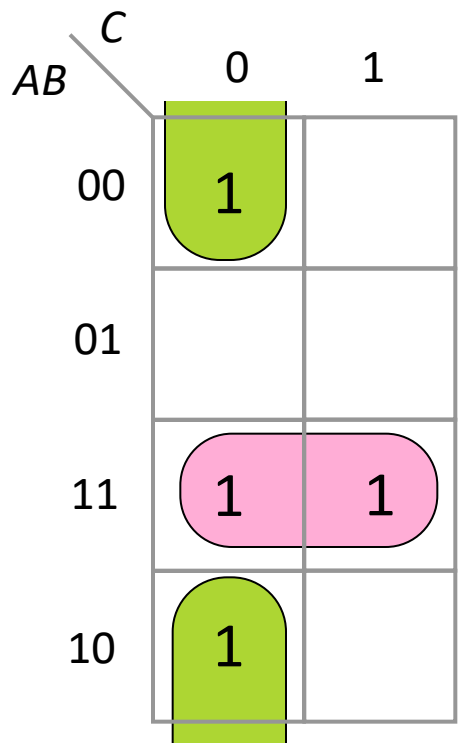
AB \ C	0	1
00	1	1
01		1
11	1	1
10	1	1

AB \ CD	00	01	11	10
00	1	1	1	
01	1	1		1
11		1		1
10		1		

$$A + C + \bar{B}$$

Mapping Truth table directly into Karnaugh map

Inputs			Output
A	**B**	**C**	**X**
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1



$$AB + \overline{B}\,\overline{C}$$

“Don’t care” conditions

- Sometimes a situation arises in which some input variable combinations are not allowed. In that case whether the value of the expression is “1” or “0” makes no difference since these combinations of inputs will never occur.
- In this case, the “don’t care” term is identified by “X” in the truth table and Karnaugh map.

A	B	C	D	Output
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

“Don’t care” conditions

- In this case of “don’t care” condition, one can use X to represent 1 or 0 in K-map.
- Result using “Xs”: $A + BCD$ not using them: $A\bar{B}\bar{C} + \bar{A}BCD$

AB \ CD	00	01	11	10
00				
01			1	
11	X	X	X	X
10	1	1	X	X

AB \ CD	00	01	11	10
00				
01			1	
11	X	X	X	X
10	1	1	X	X

Mapping POS expression on a Karnaugh map

(Try deriving the POS from the Karnaugh map, then convert it to the equivalent SOP and compare to the SOP derived from the Karnaugh map!!)

$$(A+B+C)(A+\overline{B}+C)(\overline{A}+\overline{B}+C)(\overline{A}+B+\overline{C})$$

000 010 110 101

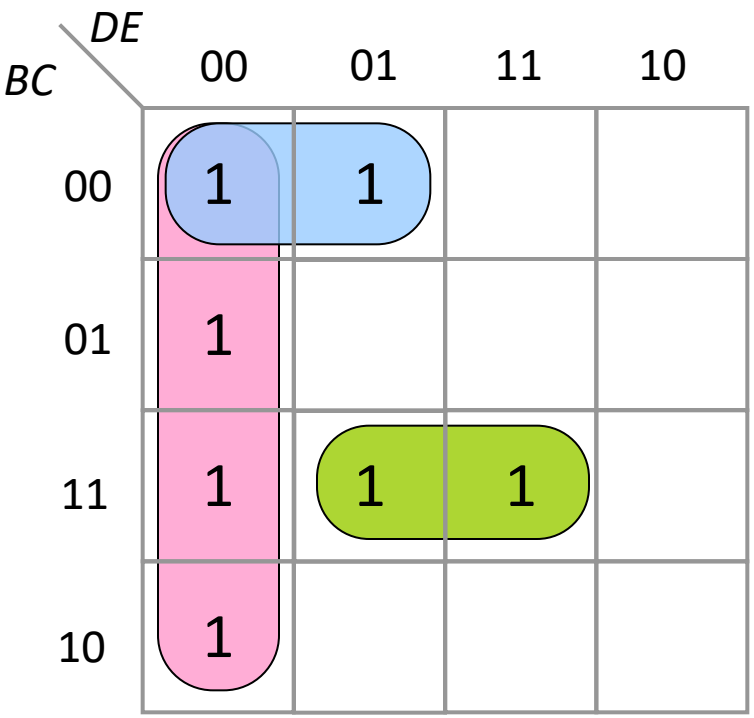
AB \ C	0	1
00	0	
01	0	
11	0	
10		0

POS and Karnaugh maps

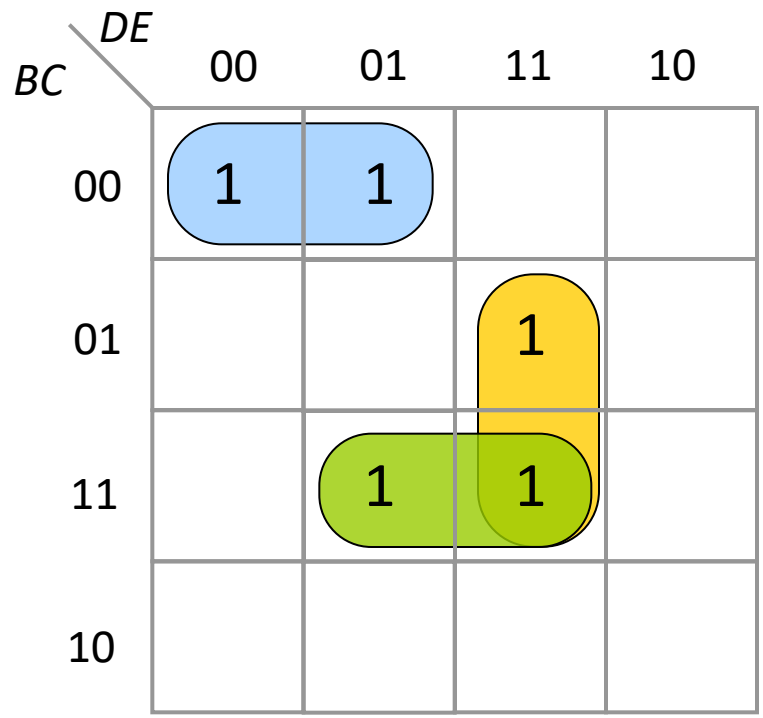
- POS expressions are treated in exactly the same way as the SOPs in connection to Karnaugh maps. **Except that in case of POS you work with 0's** rather than 1's.
- The same tricks for grouping apply. So the reduction of a POS could be done in exactly the same way as SOP, but variables have a reverse complementarity.
- Same rules for "don't care" cases apply.

5-variable Karnaugh map

- Variables domain is A, B, C, D, E
- This is the case of layering the Karnaugh map with the bottom layer signifying A=0 and top layer A=1.
- The rules and properties of such maps are still the same.



A = 0



A = 1

Class Exercise: Minimise the following using a Karnaugh map

$$AB + A\overline{B}C + ABC$$

AB \ C	0	1
00		
01		
11		
10		

$$A\overline{B}\,\overline{C}D + AC\overline{D} + B\overline{C}D + \overline{A}BC\overline{D}$$

AB \ CD	00	01	11	10
00				
01				
11				
10				

Summary

- Introduced Karnaugh maps for 3, 4 and 5 variables
- Mapped SOP and POS onto Karnaugh maps
- Used Karnaugh maps to minimise SOPs by means of grouping
- Mapped truth tables directly into Karnaugh maps
- Described a “Don’t care” condition